

Aayush Kumar

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Research Interests

I am a first-year PhD student at MIT investigating and building AI tools to improve user productivity and experiences in the workplace. By applying techniques from human-computer interaction (HCI) and AI research, I aim to develop interaction and design principles for AI systems that augment workers' capabilities while also retaining individual values, creativity and upskilling in the process.

Education

2026 –  **Ph.D. in Computer Science, MIT**, Cambridge, MA, USA
Advisor: Prof. Zana Bucinca

2020 – 2024  **B.Tech. in Computer Science and Engineering**, IIT Kanpur, India CGPA: 9.81/10
Class Rank: 8/1157

Employment History

2024-26  **Research Fellow**, PROSE team, Microsoft, India.

- Led and contributed to multiple **published research projects** as well to **product deployment** for **human-AI collaborations** in VSCode Copilot and Excel as a pre-doctoral researcher.
- Designed and implemented a **high-fidelity prototype for Plan Mode in Excel Agent**, and worked with product management and engineering teams to deploy Plan Mode to all Excel users.
- Built intuitive and interactive harnesses and **interfaces for AI agents**, conducted **user studies and mixed-methods analyses** across papers published in premier HCI and AI venues.

2023  **Quantitative Research Intern**, Quadeye Securities LLP, India.

- Performed **statistical modeling and in-depth quantitative analysis** using R to determine the effects of news sentiment data on intraday and overnight stock prices in India and the US.

Publications

Conference Papers

-  **Honorable Mention (Best Paper Runner-Up)**
A. Kumar, A. Dutta, S. Gulwani, G. Soares, A. Sarkar, and E. Murphy-Hill, "Plans Work in Mysterious Ways: Evaluating A Plan Mode for Spreadsheet Agents," in *IEEE VL/HCC*, 2026.  PRE-PRINT:
<https://arxiv.org/abs/2607.23670>
- A. Kumar, Y. Bajpai, S. Gulwani, G. Soares, and E. Murphy-Hill, "Why AI Agents Still Need You: Findings from Developer-Agent Collaborations in the Wild," in *IEEE/ACM ASE*, 2025.  DOI:
10.1109/ASE63991.2025.00043.
- J. T. Liang, A. Kumar, Y. Bajpai, S. Gulwani, V. Le, C. Parnin, A. Radhakrishna, A. Tiwari, E. Murphy-Hill, and G. Soares, "TableTalk: Scaffolding Spreadsheet Development with a Language Agent," *ACM TOCHI*, 2025.  DOI: 10.1145/3765286.
- N. Mehrotra, A. Kumar, S. Gulwani, A. Radhakrishna, and A. Tiwari, "TEN: Table Explicitization, Neurosymbolically," in *ACL (Industry Track)*, 2026.  DOI: 10.18653/v1/2026.acl-industry.138.
- H. Goel, A. Kumar, and S. S. Ragavan, "End-User Programming is WEIRD: How, Why and What to Do About It," in *IEEE VL/HCC*, 2023.  DOI: 10.1109/VL-HCC57772.2023.00013.
- M. Amoozadeh, D. Daniels, D. Nam, A. Kumar, S. Chen, M. Hilton, S. Srinivasa Ragavan, and M. A. Alipour, "Trust in Generative AI among Students: An exploratory study," in *ACM SIGCSE TS*, 2024.  DOI: 10.1145/3626252.3630842.

Workshop Posters

- 1 **A. Kumar**, "Towards Operationalizable Principles for Design Friction in Human-AI Collaborations," in *Human-AI Complementarity for Decision Making Workshop*, 2026. [🔗 POSTER:](https://aeyush10.github.io/CV/design_friction_poster.html)
https://aeyush10.github.io/CV/design_friction_poster.html

Pre-prints

- 1 **A. Kumar**, D. Prol, A. Alipour, and S. S. Ragavan, *To Google or To ChatGPT? A Comparison of CS2 Students' Information Gathering Approaches and Outcomes*, 2025. [🔗 URL:](https://arxiv.org/abs/2501.11935)
<https://arxiv.org/abs/2501.11935>
- 2 **A. Kumar** and S. Mhatre, *UnWEIRDing LLM Entity Recommendations*, 2025. [🔗 URL:](https://arxiv.org/abs/2511.18403)
<https://arxiv.org/abs/2511.18403>

Invited Talks

- 2026  **EuSpRIG 2026**
Keynote: Towards Collaboration-First AI for End-User Programming
Discussed important future research directions for AI in spreadsheet programming based on my work at Microsoft at the premier European conference for spreadsheet risks.

Fellowships, Awards and Achievements

- 2026  **SERC Scholar**, MIT.
Selected for the 2026-27 cohort of the SERC scholar program at MIT, working with a multidisciplinary team on the social and ethical considerations of AI.
-  **EECS Distinguished Science and Engineering Fellowship**, MIT.
Awarded to newly admitted PhD students who have demonstrated exemplary academic and research achievements.
- 2025  **Outstanding Reviewer**, iTiCSE 2025.
- 2024  **Research Proficiency Medal**, IIT Kanpur, India.
Awarded for best undergraduate research project by a graduating student in the Computer Science and Engineering department.
- 2022  **Academic Excellence Award**, IIT Kanpur, India.
- 2021  **Dr. Prateek Mishra Memorial Scholarship Awardee**, IIT Kanpur, India.
Awarded for achieving the highest GPA in my department during freshman year.
- 2020  **All India Rank 976**, Joint Entrance Examination - Advanced, India.
-  **All India Rank 4128**, Joint Entrance Examination - Mains, India.

Reviewing and Teaching Experience

- PC Member  VL/HCC 2026; iTiCSE 2026; BiAlign Workshop on Human-AI Interaction Alignment at NeurIPS 2026 and CHI 2026
- Reviewer  TEI 2026 (external); iTiCSE 2025; ASE 2025 (sub-reviewer); ICSE 2024 (sub-reviewer)
- Tutor  **Fundamentals of Computing**, IIT Kanpur, India.
Taught a CS1 course to a class of 25-30 students once a week during both the Fall and Spring semester in my senior year.

Selected Projects

- 1 **Intersectionality in Oversampling on Social Attributes**, Semester Project, ML4ED Lab, EPFL, 2023, Built on the paper by Cock et al. to analyze the impact of intersectionality on different oversampling strategies for a behavioural predictive model in education.
- 2 **SoulFull: An AI-Powered Companion App for Binge Eating**, Course Project, EPFL, 2023, Designed a high-fidelity prototype for an app that supports people struggling with binge eating via journalling and AI-powered insights. [🔗 URL:](https://www.figma.com/file/OLqV3BIDvyHij9iheE4esL/) <https://www.figma.com/file/OLqV3BIDvyHij9iheE4esL/>